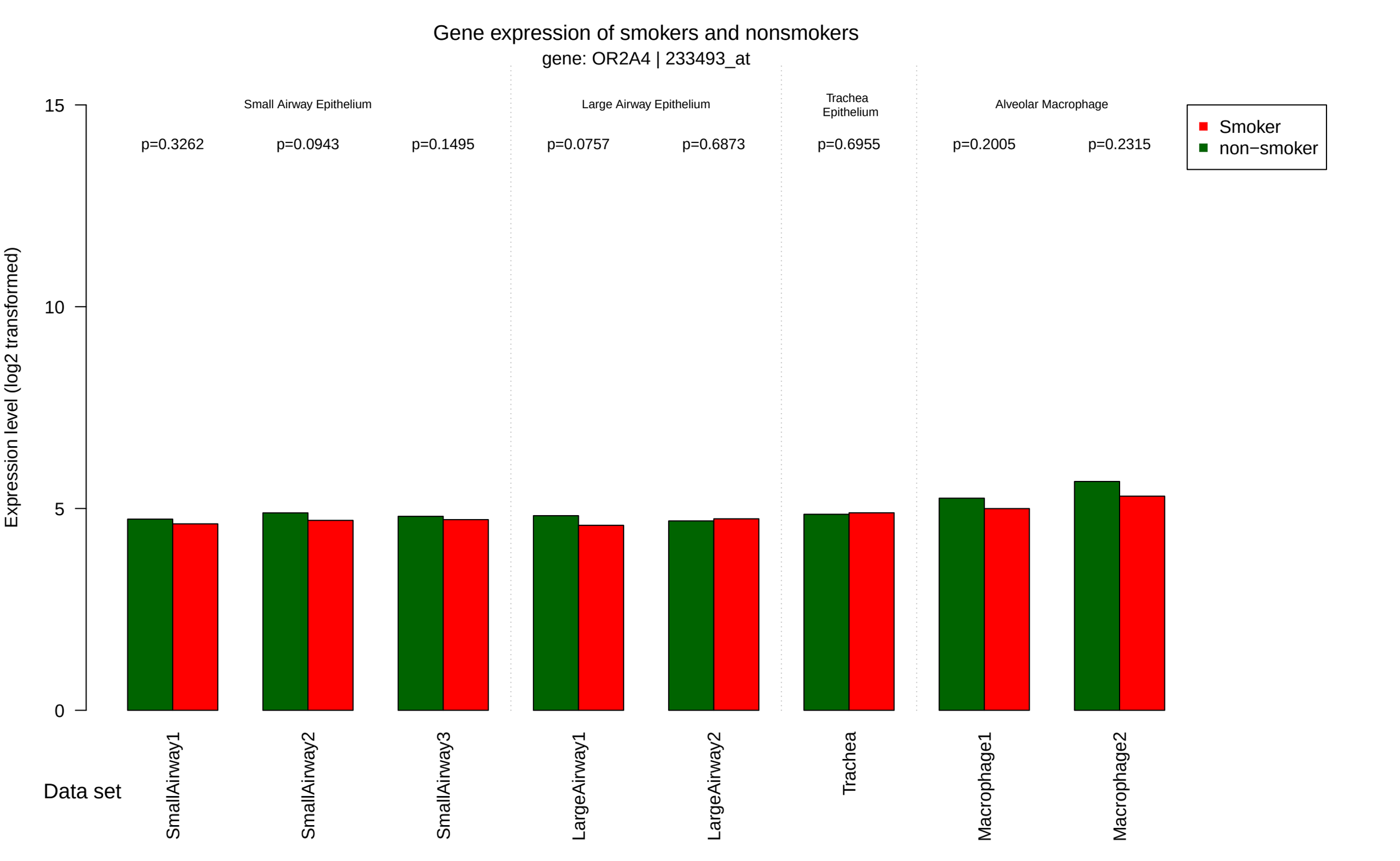
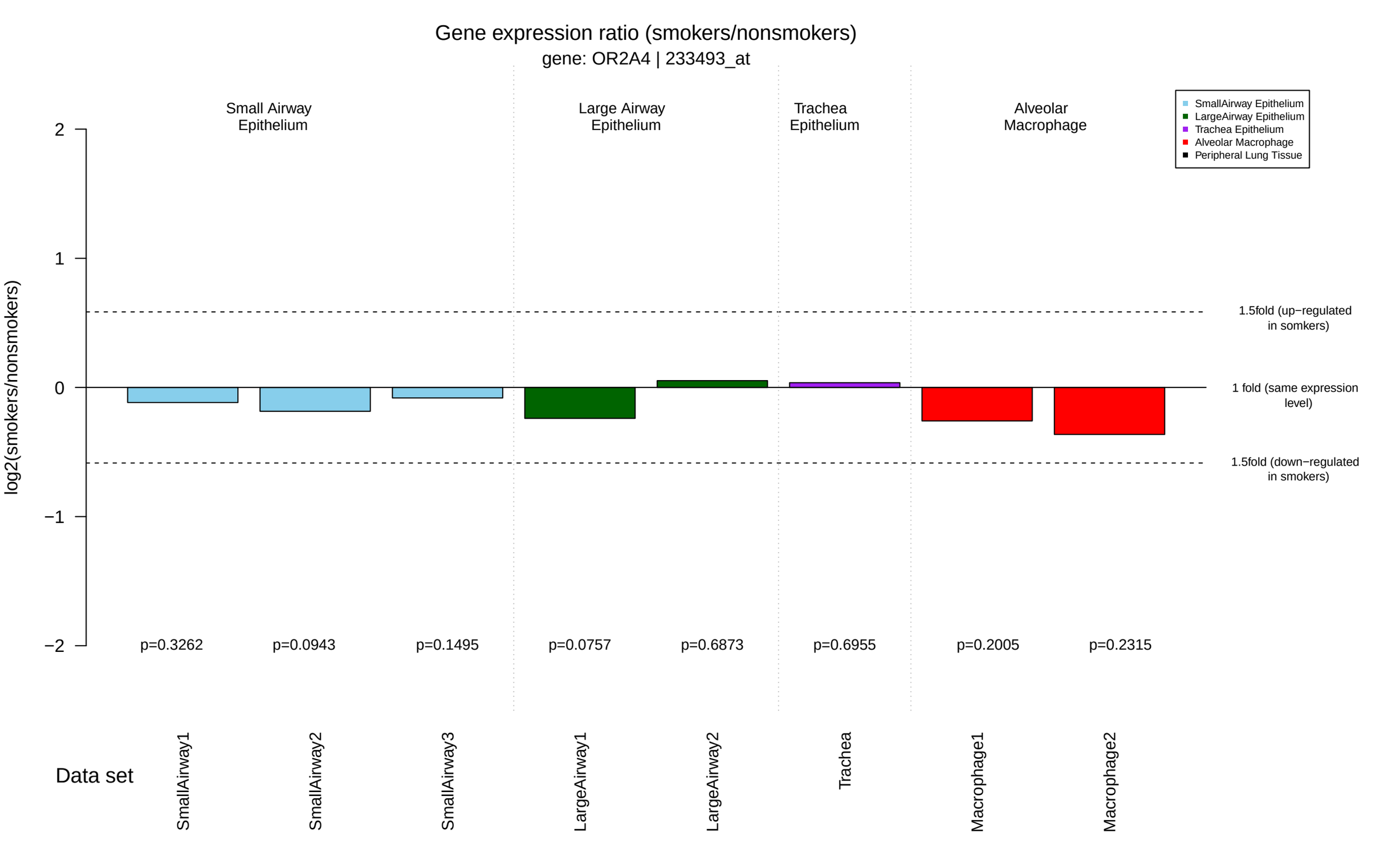
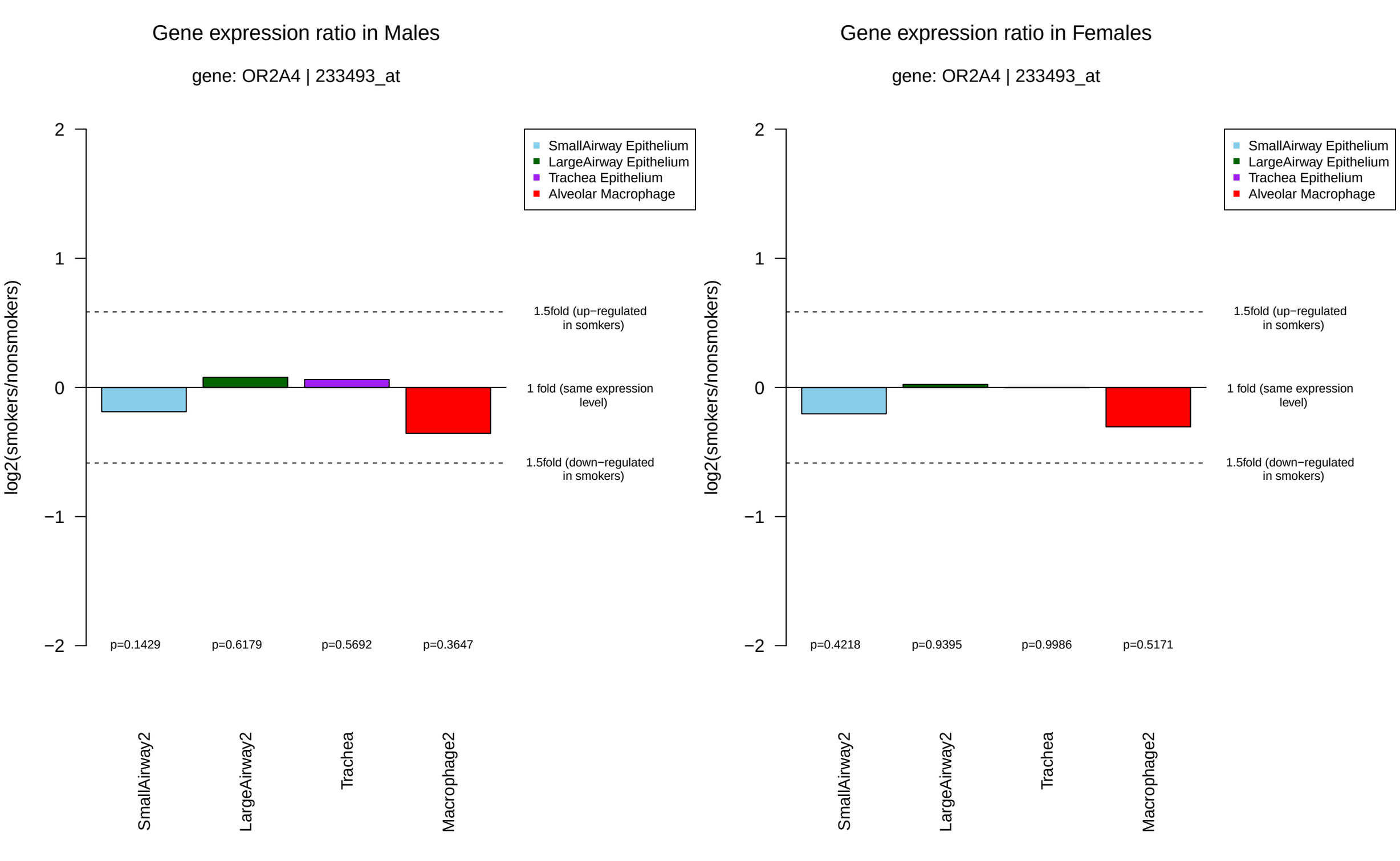
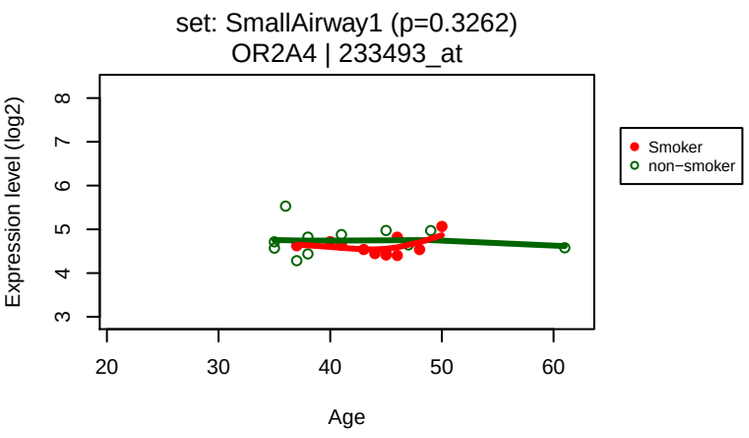
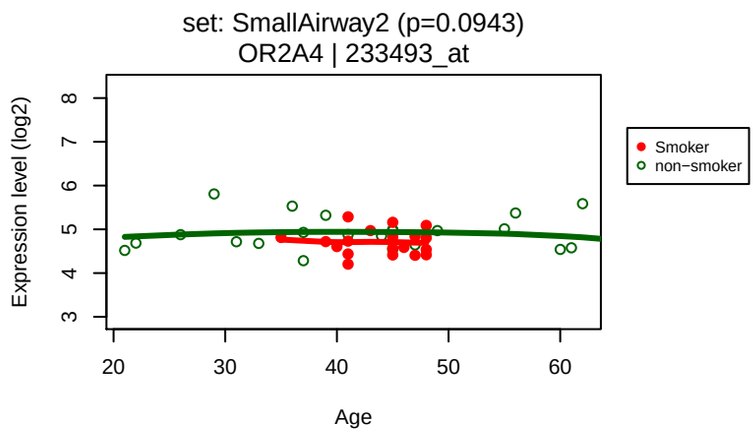
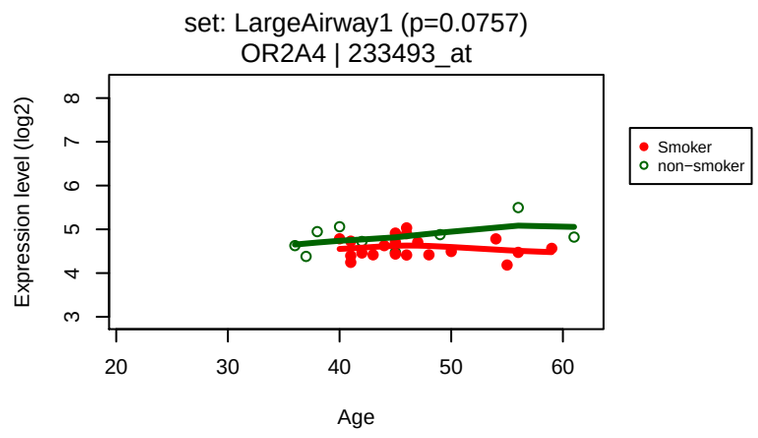

Gene expression of smokers and nonsmokers
gene: OR2A4 | 233493_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3262
p=0.0943
p=0.1495
p=0.0757
p=0.6873
p=0.6955
p=0.2005
p=0.2315
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

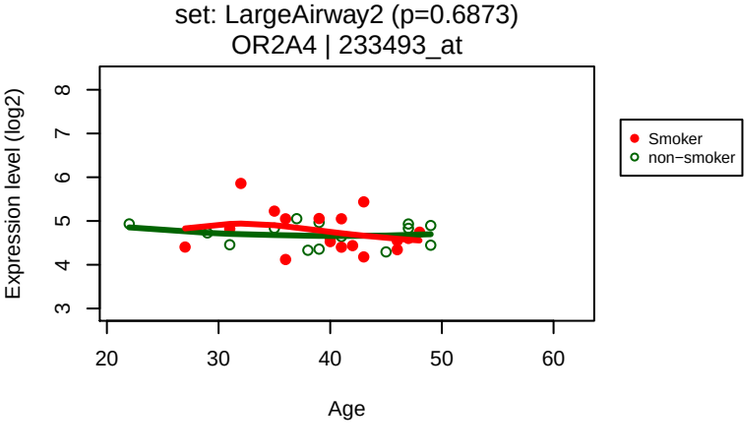

Gene expression ratio (smokers/nonsmokers)
gene: OR2A4 | 233493_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3262
p=0.0943
p=0.1495
p=0.0757
p=0.6873
p=0.6955
p=0.2005
p=0.2315
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

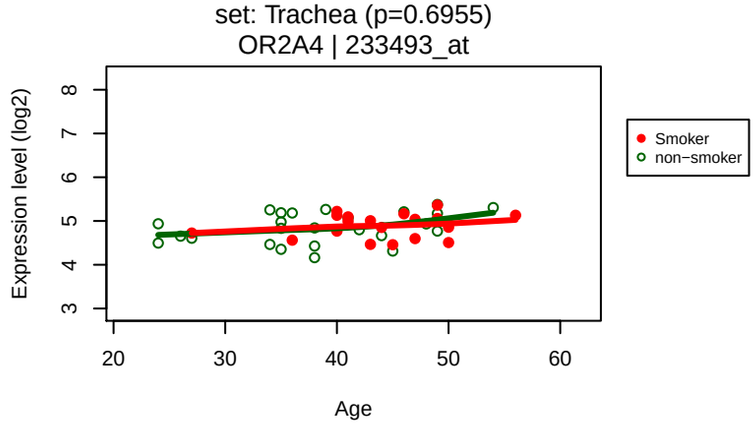

Gene expression ratio in Males
gene: OR2A4 | 233493_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1429
p=0.6179
p=0.5692
p=0.3647
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR2A4 | 233493_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4218
p=0.9395
p=0.9986
p=0.5171
SmallAirway2
LargeAirway2
Trachea
Macrophage2

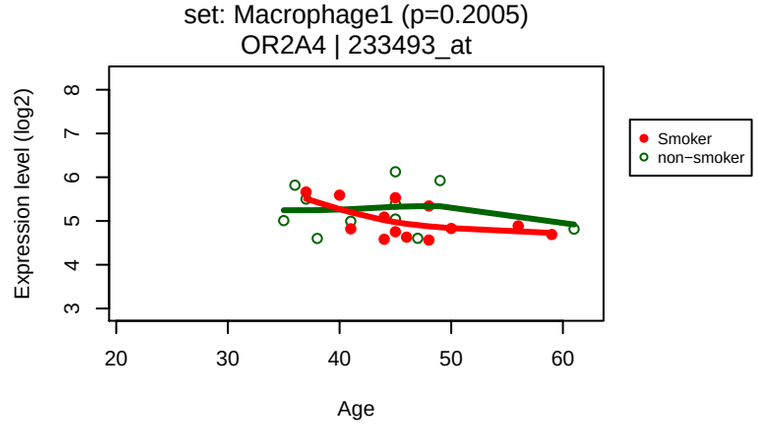

set: SmallAirway1 (p=0.3262)
OR2A4 | 233493_at
Expression level (log2)
Age
Smoker
non−smoker

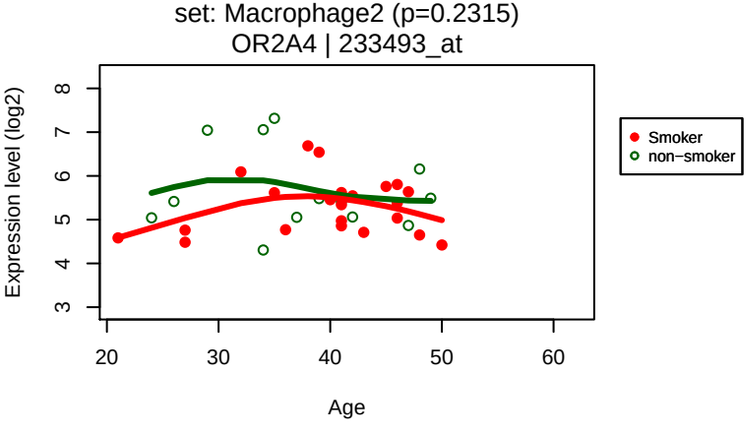

set: SmallAirway2 (p=0.0943)
OR2A4 | 233493_at
Expression level (log2)
Age
Smoker
non−smoker

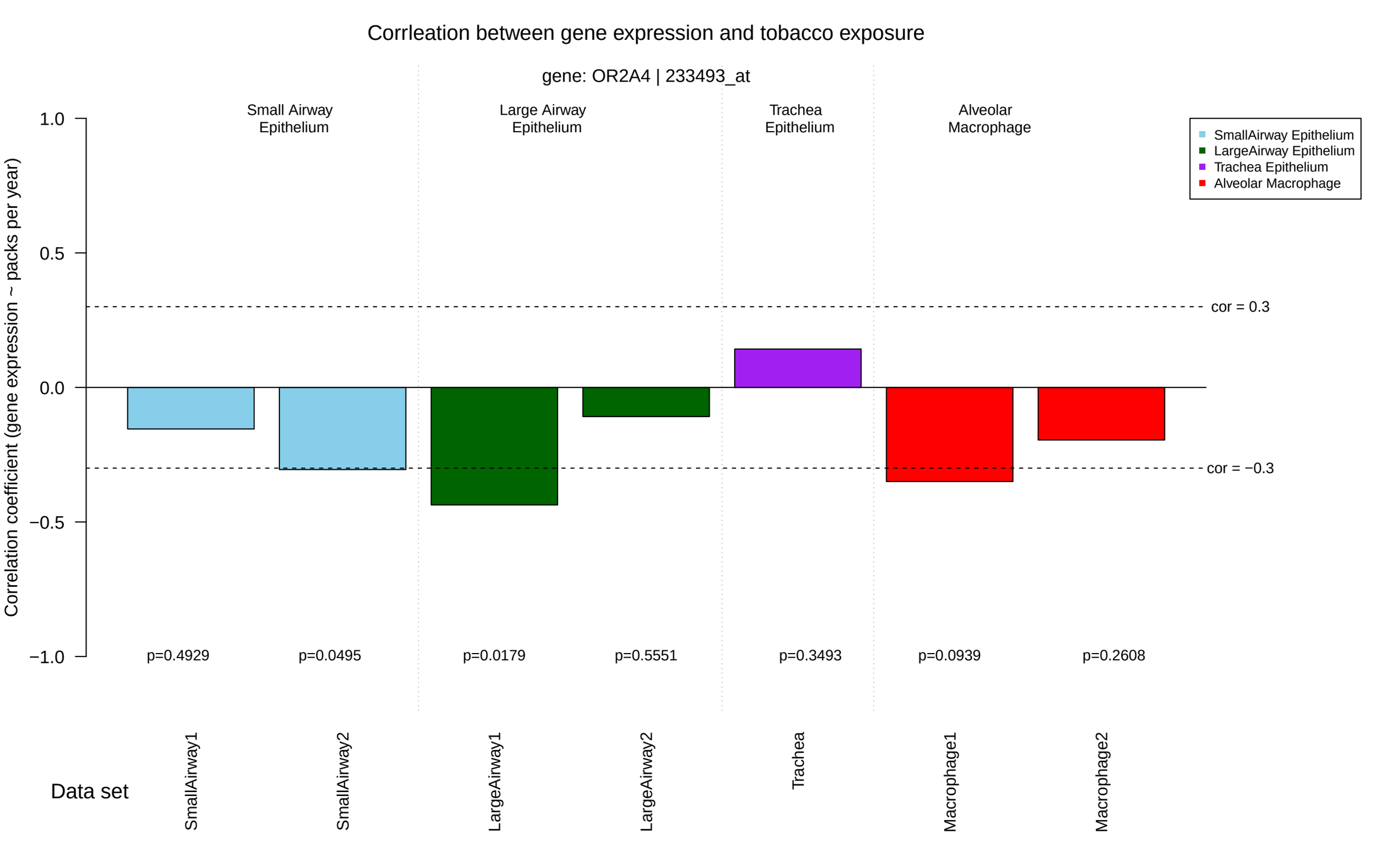

set: LargeAirway1 (p=0.0757)
OR2A4 | 233493_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6873)
OR2A4 | 233493_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6955)
OR2A4 | 233493_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2005)
OR2A4 | 233493_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2315)
OR2A4 | 233493_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR2A4 | 233493_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4929
p=0.0495
p=0.0179
p=0.5551
p=0.3493
p=0.0939
p=0.2608
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2